

# QC: Week 3

Rudraksh Goyal

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## Why the Gate Count Grows So Rapidly

The aggressive explosion in gate count as error tolerance ( $\epsilon$ ) decreases is driven by three main factors:

- **Polynomial Single-Angle Scaling ( $O(1/\epsilon)$ ):** The `approx_angle_with_tolerance` function searches for an integer  $k$  such that  $k\lambda_\pi \approx \text{angle} \pmod{2\pi}$ . By Dirichlet's Approximation Theorem, guaranteeing accuracy within  $\epsilon$  requires  $k$  to scale as  $O(1/\epsilon)$ , causing a steep polynomial growth in required  $H/T$  gates.
- **Structural Pipeline Cascading:** Decomposing a  $4 \times 4$  unitary fractures the matrix into two-level unitaries, Gray-code multi-controlled gates, and single-qubit ZYZ Euler rotations. Since every single-qubit gate requires three independent angles that must each meet the target tolerance  $\epsilon$ , the  $O(1/\epsilon)$  blowup multiplies heavily across the compiled circuit.
- **Lack of Logarithmic Optimization:** Professional compilers leverage advanced algorithms like the **Solovay-Kitaev Theorem** to achieve efficient polylogarithmic scaling ( $O(\log^c(1/\epsilon))$ ). Because this pipeline relies on a brute-force sequential search, it scales strictly as  $O(1/\epsilon)$ , forcing gate counts to skyrocket under tighter error thresholds.